Class 8 Mini-Project

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Background

In today's class, we will apply the methods and techniques clustering and PCA to help make sense of a real world breast cancer FNA biopsy data set.

Preparing the Data

Start by importing our data. It is a CSV file so we will use the `read.csv()` function.

```
# Save under "temp_data" so entire csv file does not render
temp_data <- read.csv("WisconsinCancer.csv")
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

wisc.df <- read.csv(fna.data, row.names=1)
```

Make sure to remove the first `diagnosis` column - I don't want to use this for my machine learning models. We will use it later to compare our results to the expert diagnosis.

```
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```
sum(wisc.df$diagnosis == "M") + sum(wisc.df$diagnosis == "B")
```

```
[1] 569
```

```r
# Another way to answer this question
length(wisc.df$diagnosis)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```r
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

```r
# Another way to answer this question
table(wisc.df$diagnosis)
```

```
  B   M 
357 212 
```

Q3. How many variables/features in the data are suffixed with _mean?

```r
colnames(wisc.df)[grepl("_mean$", colnames(wisc.df))]
```

```
 [1] "radius_mean"            "texture_mean"           "perimeter_mean"        
 [4] "area_mean"              "smoothness_mean"        "compactness_mean"      
 [7] "concavity_mean"         "concave.points_mean"    "symmetry_mean"         
[10] "fractal_dimension_mean"
```

```r
length(colnames(wisc.df)[grepl("_mean$", colnames(wisc.df))])
```

```
[1] 10
```

```r
# Another way to answer this question
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

Principal Component Analysis

The main function here is `prcomp()` and we want to make sure we set the optional argument `scale = TRUE`:

```
# Check column means and standard deviations
colMeans(wisc.data)
```

```
            radius_mean           texture_mean         perimeter_mean
           1.412729e+01           1.928965e+01           9.196903e+01
              area_mean        smoothness_mean       compactness_mean
           6.548891e+02           9.636028e-02           1.043410e-01
         concavity_mean    concave.points_mean          symmetry_mean
           8.879932e-02           4.891915e-02           1.811619e-01
 fractal_dimension_mean              radius_se             texture_se
           6.279761e-02           4.051721e-01           1.216853e+00
           perimeter_se                area_se          smoothness_se
           2.866059e+00           4.033708e+01           7.040979e-03
         compactness_se           concavity_se      concave.points_se
           2.547814e-02           3.189372e-02           1.179614e-02
            symmetry_se   fractal_dimension_se           radius_worst
           2.054230e-02           3.794904e-03           1.626919e+01
          texture_worst        perimeter_worst             area_worst
           2.567722e+01           1.072612e+02           8.805831e+02
       smoothness_worst      compactness_worst        concavity_worst
           1.323686e-01           2.542650e-01           2.721885e-01
   concave.points_worst         symmetry_worst fractal_dimension_worst
           1.146062e-01           2.900756e-01           8.394582e-02
```

```
apply(wisc.data,2,sd)
```

```
            radius_mean           texture_mean         perimeter_mean
           3.524049e+00           4.301036e+00           2.429898e+01
              area_mean        smoothness_mean       compactness_mean
           3.519141e+02           1.406413e-02           5.281276e-02
         concavity_mean    concave.points_mean          symmetry_mean
           7.971981e-02           3.880284e-02           2.741428e-02
 fractal_dimension_mean              radius_se             texture_se
           7.060363e-03           2.773127e-01           5.516484e-01
           perimeter_se                area_se          smoothness_se
           2.021855e+00           4.549101e+01           3.002518e-03
```

```
        compactness_se            concavity_se       concave.points_se
          1.790818e-02            3.018606e-02            6.170285e-03
           symmetry_se    fractal_dimension_se            radius_worst
          8.266372e-03            2.646071e-03            4.833242e+00
         texture_worst         perimeter_worst              area_worst
          6.146258e+00            3.360254e+01            5.693570e+02
      smoothness_worst       compactness_worst         concavity_worst
          2.283243e-02            1.573365e-01            2.086243e-01
  concave.points_worst          symmetry_worst fractal_dimension_worst
          6.573234e-02            6.186747e-02            1.806127e-02
```

```r
# Perform PCA on wisc.data
wisc.pr <- prcomp(wisc.data, scale = T)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 captures 44.27% (0.4427) of the original variance.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

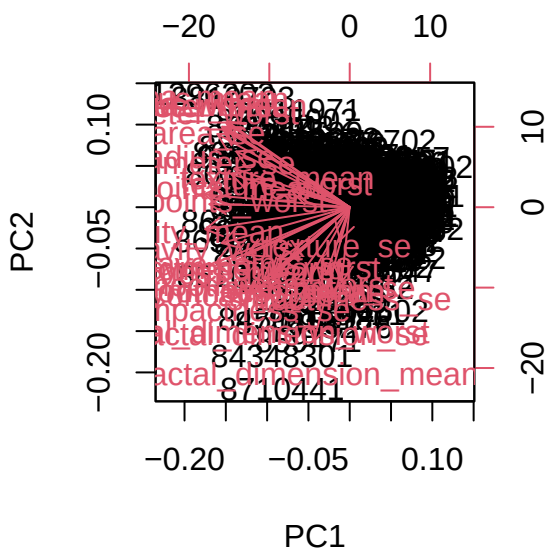
Three PCs capture 72.63% (0.72633) of the original variance.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Seven PCs capture 91.01% (0.91008) of the original variance.

Interpreting PCA Results

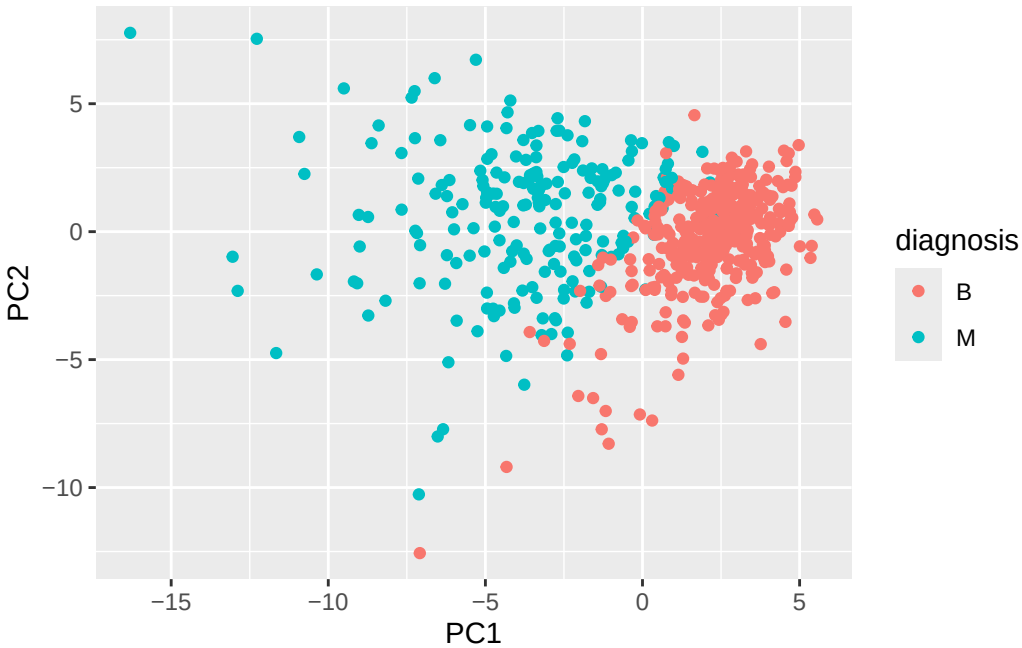
```
biplot(wisc.pr)
```



> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

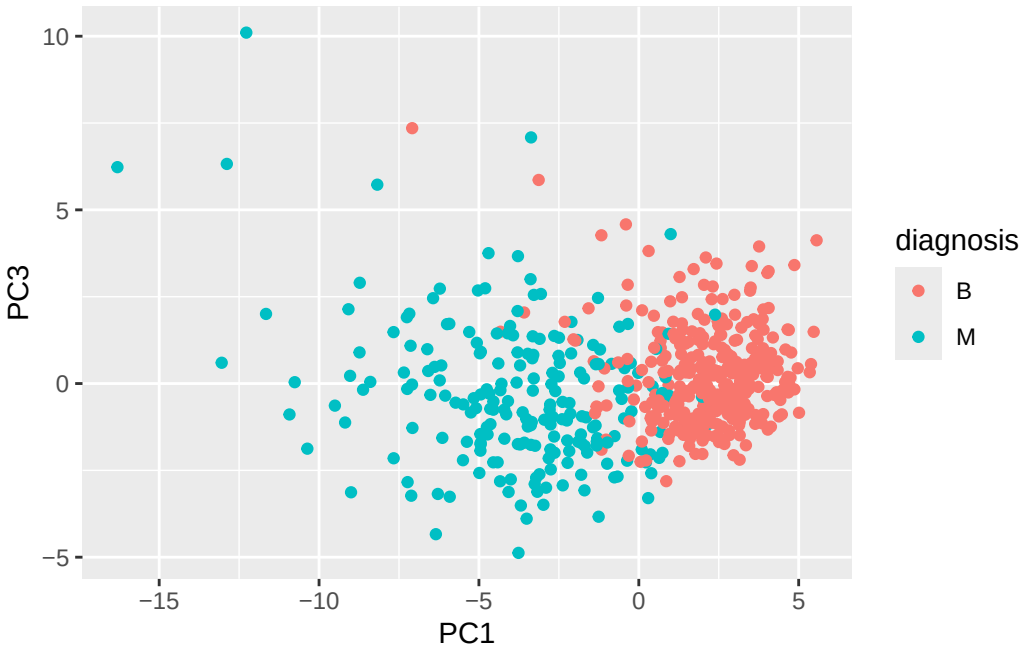
The plot above is clustered around the center, (0, 0). The plot is extremely difficult to understand and interpret because the points and labels are overlapping on top of each other and the viewer cannot read them.

```r
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```r
# Repeat for components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



The plots show that PC1 is capturing a separation of malignant (turquoise, "M") from benign (red, "B") samples.

This is an important and interesting result worthy of further exploration - as we will do in the next sections.

Communicating PCA Results

> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

The component of the loading vector of PC1 is -0.2608538 for the feature concave.points_mean, and indicates a strong negative contribution to the first principal component. Other features like concave.points_worst, concavity_mean, or area_worst have slightly larger absolute loading values.

Heirarchial Clustering

Conduct hierarchical clustering of the original data to see if there are any obvious groupings into malignant and benign clusters.

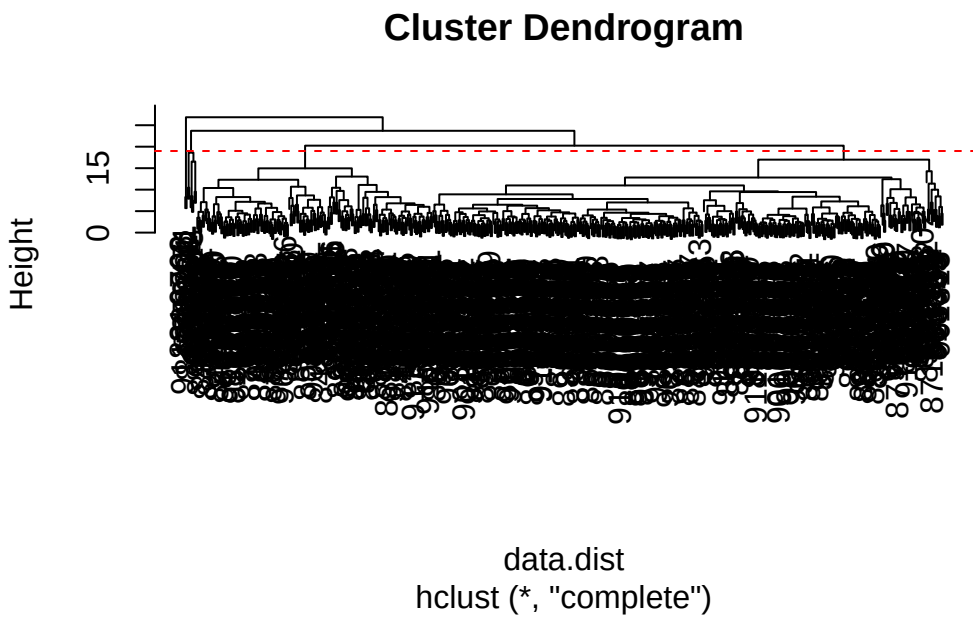
```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)

# Calculate the (Euclidean) distances between all pairs of observations in the new scaled dat
data.dist <- dist(data.scaled)

# Create a hierarchical clustering model using complete linkage. Manually specify the method
wisc.hclust <- hclust(data.dist, method = "complete")
wisc.hclust
```

```
Call:
hclust(d = data.dist, method = "complete")

Cluster method   : complete
Distance         : euclidean
Number of objects: 569
```

Results of Hierarchical Clustering

> Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



The height at which the clustering model has 4 cluster is h=19.

Selecting Number of Clusters

Let's see how well the clustering separates malignant from benign samples. You can use `cutree()` function with an argument `k=4` rather than `h=height`.

```
# Use cutree() to cut the tree so that it has 4 clusters
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)

# Use the table() function to compare the cluster membership to the actual diagnoses
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1   2   3   4
177   7 383   2
```

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
```

```
1   12 165
2    2   5
3  343  40
4    0   2
```

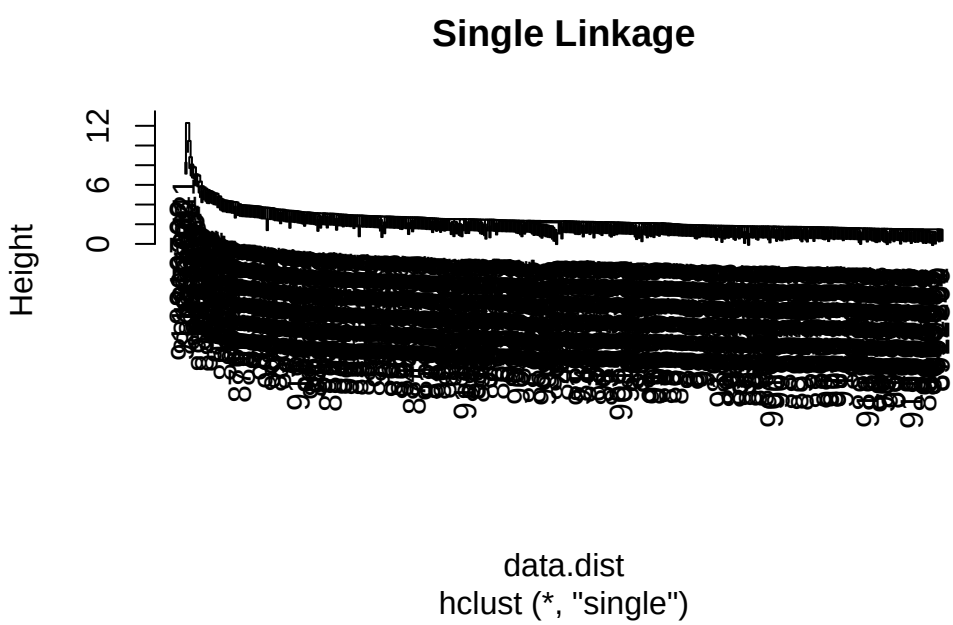
Q11. Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case? (OPTIONAL)

Using Different Methods

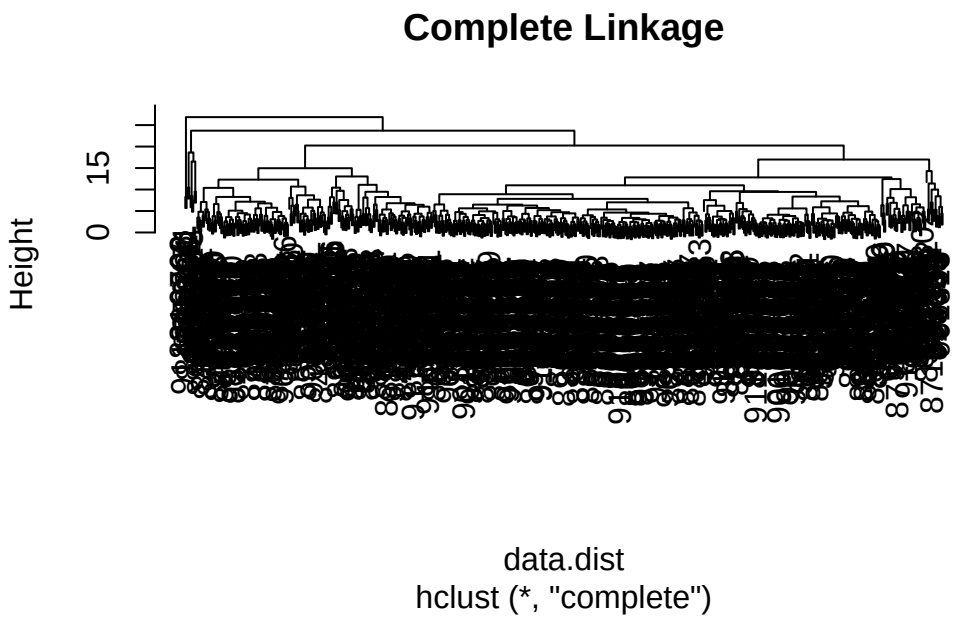
There are number of different "methods" we can use to combine points during the hierarchical clustering procedure. These include "single", "complete", "average" and "ward.D2"

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

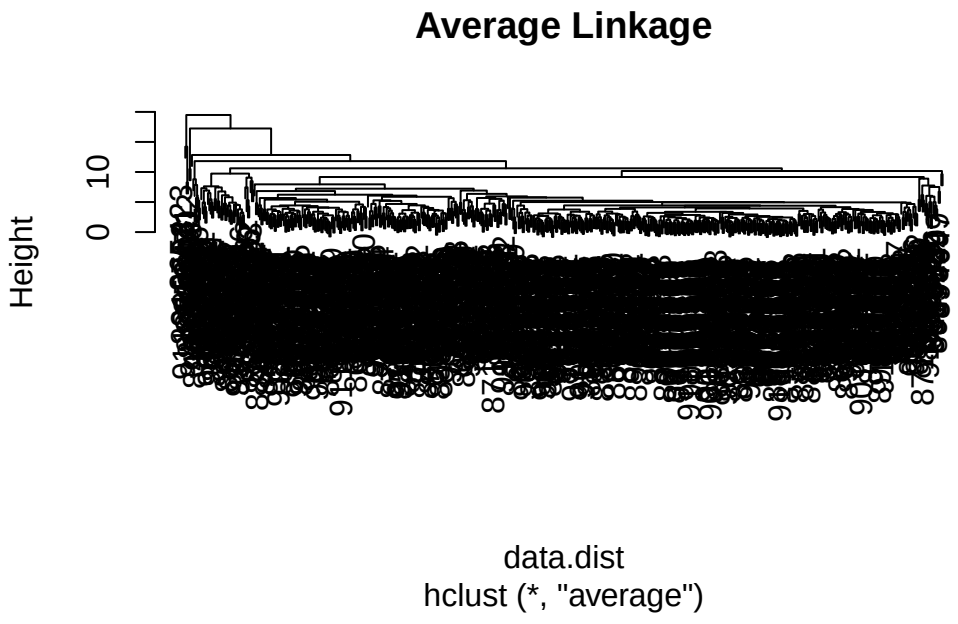
```r
hc.single <- hclust(data.dist, method = "single")
hc.complete <- hclust(data.dist, method = "complete")
hc.average <- hclust(data.dist, method = "average")
hc.ward <- hclust(data.dist, method = "ward.D2")

plot(hc.single, main="Single Linkage")
```



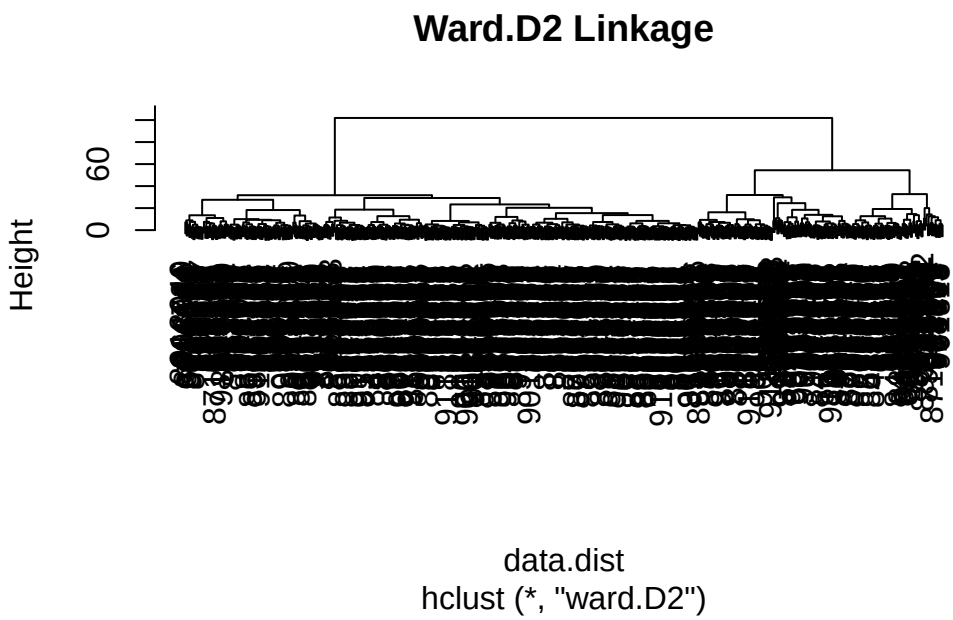
```r
plot(hc.complete, main="Complete Linkage")
```




```r
plot(hc.average, main="Average Linkage")
```



```r
plot(hc.ward, main="Ward.D2 Linkage")
```



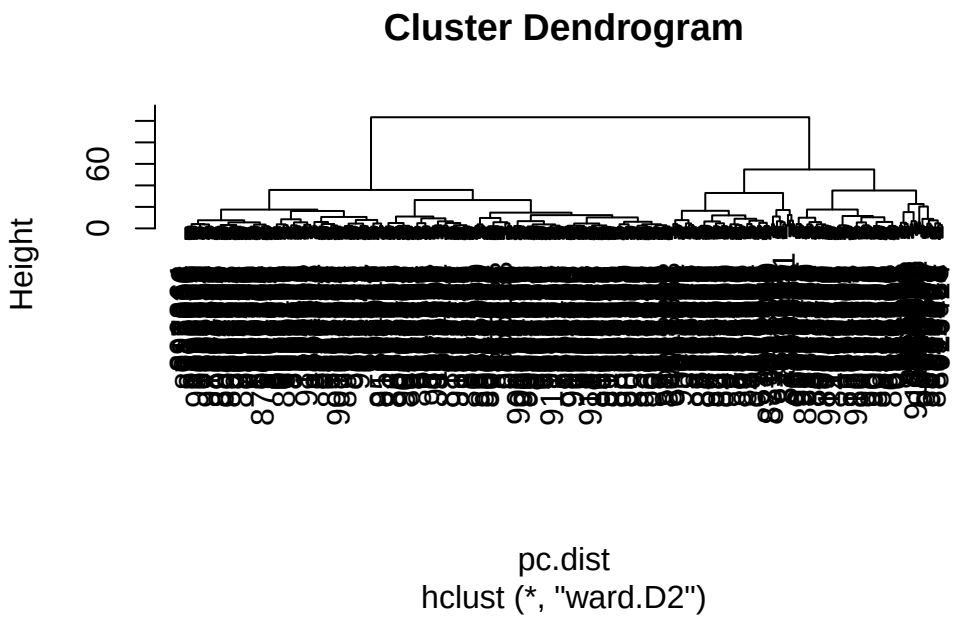
"ward.D2" method gives the best result for hierarchical clustering because it minimizes within-cluster variance, leading to more compact and well-separated clusters that can better reflect underlying biological groups in datasets.

Combining Methods (Clustering on PCA Results)

Here we will take our PCA results and use those as input for clustering. In other words, our `wisc.pr$x` scores that we plotted above (the main output from PCA - how data lie on our new principal component axis/variables) and use a subset of the PCs that capture the most variance as input for `hclust()`.

Let's see if PCA improves or degrades the performance of hierarchical clustering. Create a hierarchical clustering model with the linkage method="ward.D2". We use Ward's criterion here because it is based on multidimensional variance like principal components analysis.

```r
pc.dist <- dist(wisc.pr$x[, 1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```

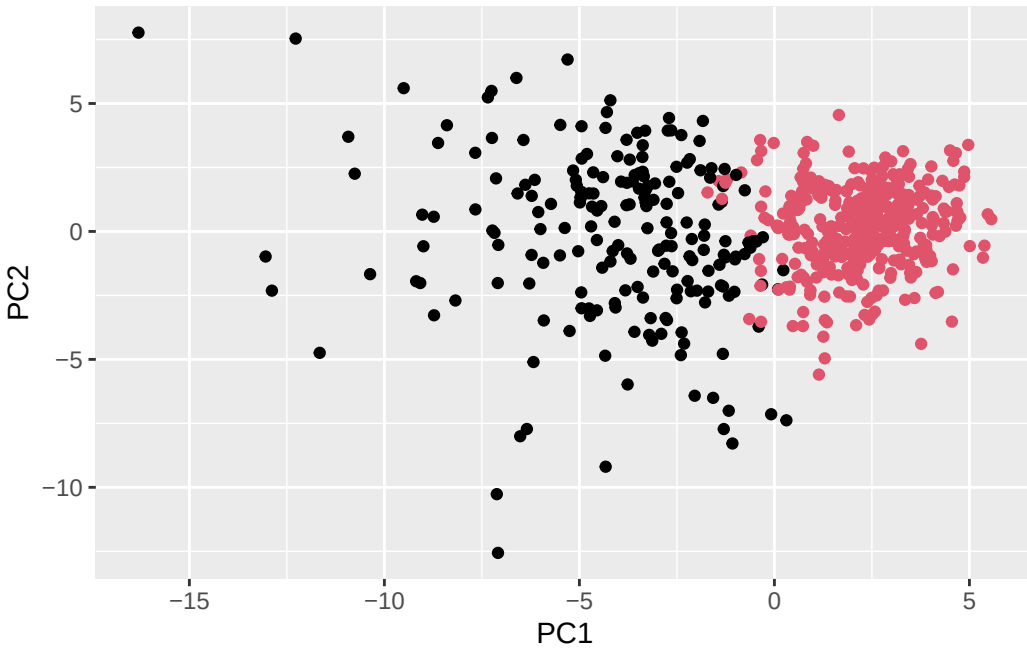


Cut the dendrogram/tree into two main groups/clusters.

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



I want to know how the clustering in `grps` with values of 1 or 2 correspond to the expert diagnosis.

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

My clustering **group 1** are mostly "M" diagnosis (179) and my clustering **group 2** are mostly "B" diagnosis (24).

There are 24 false positives and 179 true positives. There are 333 true negatives and 33 false negatives.

24 FP 179 TP 333 TN 33 FN

```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method="ward.D2")
```

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The newly created hclust model with two clusters separates "M" and "B" well. Although the separation is not 100%, the majority of "M" and "B" will be grouped together in their corresponding cluster. The "ward.D2" linkage method works to create distinct clusters by minimizing within-cluster variance, improving the cluster separation.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)

table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

The PCA-based model performed better in separating the diagnoses. By comparing the two tables, the PCA table has fewer misclassified samples with smaller numbers in the off-diagonal cells. The model that used the raw data, without PCA, has more "M" samples mistakenly grouped with "B" samples, and vice versa.

Sensitivity/Specificty

> Q15. Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity? (OPTIONAL)

Sensitivity TP/(TP+FN)

```r
179/(179+33)
```

```
[1] 0.8443396
```

Specificity TN/(TN+FP)

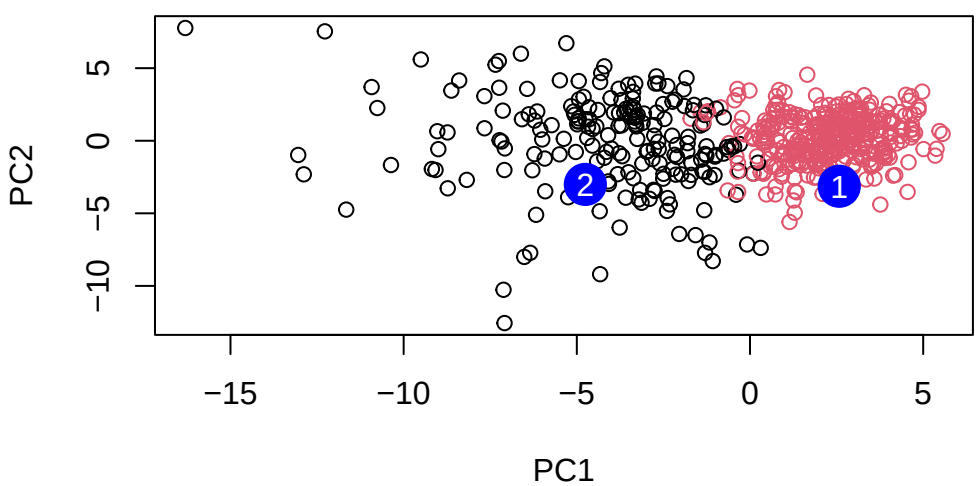
```r
333/(333+24)
```

```
[1] 0.9327731
```

Prediction

Use `predict()` to take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```r
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)

plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



> Q16. Which of these new patients should we prioritize for follow up based on your results?

Based on the results, the patient that should be prioritized for follow-up are those whose blue data points land within the "M" cluster region of the plot. Patients with characteristics aligning with malignant tumors are encouraged to get immediate clinical attention.

Summary and About this Document

These methods/approaches are a great way to analyze complex data.

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: x86_64-apple-darwin20
Running under: macOS Sequoia 15.7.3

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] ggplot2_4.0.1

loaded via a namespace (and not attached):
 [1] vctrs_0.6.5        cli_3.6.5          knitr_1.51         rlang_1.1.7
 [5] xfun_0.55          generics_0.1.4     S7_0.2.1           jsonlite_2.0.0
 [9] labeling_0.4.3     glue_1.8.0         htmltools_0.5.9    scales_1.4.0
[13] rmarkdown_2.30     grid_4.5.2         evaluate_1.0.5     tibble_3.3.1
[17] fastmap_1.2.0      yaml_2.3.12        lifecycle_1.0.5    compiler_4.5.2
[21] dplyr_1.1.4        RColorBrewer_1.1-3 pkgconfig_2.0.3    farver_2.1.2
[25] digest_0.6.39      R6_2.6.1           tidyselect_1.2.1   pillar_1.11.1
[29] magrittr_2.0.4     withr_3.0.2        tools_4.5.2        gtable_0.3.6
```